SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Design Task

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Design Task

1. **Hybrid AI Recommendation Engine Design**
   Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. **Smart Disaster Detection System**
   Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. **AI-Based Fraud Detection Framework**
   Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. **Autonomous Supply Chain Optimization System**
   Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. **Personalized Learning Analytics Platform**
   Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

ANSWERS

1. Hybrid AI Recommendation Engine

Objective

Build a recommendation system combining content-based filtering and collaborative filtering using PySpark RDDs.

Architecture

User Data → PySpark RDD → Feature Processing

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Content + Collaborative Filtering

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Hybrid Recommendation

↓

Intelligent Agents

↓

User Feedback

PySpark RDD Operations

- **map() – extract user/item features.**
- **filter() – remove invalid data.**
- **reduceByKey() – calculate ratings/counts.**
- **join() – combine user and item data.**
- **sortBy() – rank recommendations.**

Hybrid score:

$$Score = \alpha(CF) + (1 - \alpha)(CB)$$

Intelligent Agents

- **User Agent: observes user behavior.**
- **Recommendation Agent: generates suggestions.**

- **Feedback Agent: analyzes likes/dislikes.**
- **Learning Agent: updates recommendations.**

Scalability

RDDs divide millions of user-item interactions across Spark worker nodes for parallel processing.

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2. Smart Disaster Detection System

Objective

Detect floods, earthquakes, etc. using satellite, sensor, weather, and social-media data.

Architecture

Satellite + Sensors + Social Media

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PySpark RDD

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Data Cleaning/Fusion

↓

AI Detection

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Intelligent Agents

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Early Warning

RDD Operations

- **map() – extract sensor/image features.**

- **filter() – remove noisy data.**
- **reduceByKey() – aggregate sensor readings.**
- **join() – combine different data sources.**

Data Fusion

Multiple sources are combined to calculate disaster confidence.

Intelligent Agents

- **Sensor monitoring agent.**
- **Image analysis agent.**
- **Social-media agent.**
- **Emergency response agent.**

Scalability

Different geographical regions can be processed in parallel using RDD partitions.

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3. AI-Based Fraud Detection

Objective

Detect suspicious financial transactions from large-scale transaction data.

Architecture

Transactions → PySpark RDD → Feature Extraction

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Anomaly Detection

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Intelligent Agents

↓

Fraud Alert

↓

Feedback

RDD Operations

```
txRDD.filter(...)
     .map(...)
     .reduceByKey(...)
     .join(...)
```

Anomaly score:

$$Z = \frac{x - \mu}{\sigma}$$

A high Z-score can indicate unusual behavior.

Intelligent Agents

- **Monitoring Agent: monitors transactions.**
- **Anomaly Agent: detects unusual patterns.**
- **Investigation Agent: examines suspicious accounts.**
- **Learning Agent: improves the fraud model.**

Scalability

Transactions are distributed across Spark nodes, allowing millions of records to be analyzed quickly.

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4. Autonomous Supply Chain Optimization

Objective

Optimize inventory, transportation, suppliers, and demand forecasting.

Architecture

Sales + Inventory + Supplier + Logistics Data

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PySpark RDD

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Demand Forecasting

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Optimization Engine

↓

Intelligent Agents

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Supply Chain Decisions

RDD Operations

- **map() – transform sales data.**
- **reduceByKey() – calculate product demand.**
- **groupByKey() – group warehouse data.**
- **join() – combine supplier and inventory data.**

Intelligent Agents

- **Inventory Agent.**
- **Supplier Agent.**
- **Logistics Agent.**
- **Demand Forecasting Agent.**

- **Coordination Agent.**

Objective Function

$$Total\ Cost = Inventory + Transport + Ordering + Shortage$$

Agents cooperate to minimize total cost.

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5. Personalized Learning Analytics

Objective

Analyze student behavior and performance to provide personalized learning.

Architecture

Student Activity

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PySpark RDD

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Performance Analysis

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Student Profile

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Recommendation Agent

↓

Personalized Content

↓

Student Feedback

RDD Operations

- **map() – process student events.**
- **filter() – remove invalid records.**
- **reduceByKey() – calculate average scores.**
- **groupByKey() – group activities by student.**

Intelligent Agents

- **Learner Agent: maintains student profile.**
- **Content Agent: recommends lessons.**
- **Assessment Agent: selects questions.**
- **Feedback Agent: provides immediate feedback.**

Adaptive Learning

If a student performs poorly, the agent provides easier explanations and additional practice. If performance improves, difficulty increases.

Rubric – Quick Points

Rubric	Key Point
Problem Understanding	Clearly identify input, problem, and objective
Architecture	Data → PySpark → AI → Agents → Decision
PySpark/RDD	map, filter, reduceByKey, join, groupByKey

Rubric	Key Point
Intelligent Agents	**Monitor, analyze, decide, learn**
Scalability	**RDD partitions enable parallel processing**
Data Flow	**Input → Cleaning → Processing → AI → Action → Feedback**
Innovation	**Real-time adaptation and multi-agent coordination**
Clarity	**Use architecture diagrams and concise explanations**

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation